


January 8, 2026

Base 2, 8, 16

$$\begin{array}{cccccccccccc} & 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 & & & & & \\ \text{Binary } (11111111) & \rightarrow & (& & & & & &) & & & & & \\ & & & & & & & & & & & & & \end{array}$$
$$1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$$
$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 =$$
$$= 255_{10}$$

$$(377)_8 = 3 \cdot 8^2 + 7 \cdot 8^1 + 7 \cdot 8^0 = 255_{10}$$

$$(FF)_{16} \rightarrow ()_{10}$$

base 16:

$$16^{\overset{1}{F}} + 16^{\circ F} = 16 \cdot 15 + 15 = 255_{10}$$

Decimal
 $()_{10} \rightarrow ()_2$
 $()_{10} \rightarrow ()_8$
 $()_{10} \rightarrow ()_{16}$

10
11
12
13
14
15

A
B
C
D
E
F

HEX

$$\begin{array}{r}
 255 \mid 2 \\
 \hline
 254 \mid 127 \mid 2 \\
 \hline
 126 \mid 63 \mid 2 \\
 \hline
 62 \mid 31 \mid 2 \\
 \hline
 30 \mid 15 \mid 2 \\
 \hline
 14 \mid 7 \mid 2 \\
 \hline
 6 \mid 3 \mid 2 \\
 \hline
 2 \mid 1 \\
 \hline
 \end{array}$$

255 = 11111111

10 2 1 1 1 1 1 1

1 1 1 1 1 1 1 1

114₁₀

$$\begin{array}{r|l} 114 & 2 \\ \hline 114 & 57 \\ 0 & 56 \\ & 128 \\ & 128 \\ & 14 \\ 0 & 14 \\ & 7 \\ & 6 \\ & 3 \\ 1 & 2 \\ & 1 \end{array}$$

1110010₂

$$(0.12)_{10} \rightarrow ()_2 = (.000111)_2$$

$0.12 \times 2 = 0.24$	0	just fraction ↓
$0.24 \times 2 = 0.48$	0	
$0.48 \times 2 = 0.96$	0	
$0.96 \times 2 = 1.92$	1	
$0.92 \times 2 = 1.84$	1	
$0.84 \times 2 = 1.68$	1	

Dec.	Binary	Dec.	Binary
1	0001	9	1001
2	0010	10	1010
3	0011	11	1011
4	0100	12	1100
5	0101	13	1101
6	0110	14	1110
7	0111	15	1111
8	1000		

A B C D E F
hex.

Binary addition

$$\begin{array}{r}
 \times \quad 01100 \\
 \quad 10001 \\
 \hline
 \quad 11101
 \end{array}$$

$$\begin{array}{r}
 \overset{1}{+} \overset{1}{1} \overset{1}{0} \overset{1}{1} 1 0 \\
 \underline{10111} \\
 1[01101]
 \end{array}$$

$$(1)_2 + (1)_2 = (10)_2$$

$$(\overset{1}{1}0)_2 + (1)_2 = (11)_2$$

It is overflow, machine should know that it is overflow

Binary subtraction

$$\begin{array}{r}
 \overset{0}{-} \overset{0}{0} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{1}{1} \\
 \hline
 0
 \end{array}
 \quad
 \begin{array}{r}
 \overset{1}{-} \overset{0}{0} \\
 \hline
 1
 \end{array}
 \quad
 \begin{array}{r}
 \overset{0}{-} \overset{1}{1} \\
 \hline
 1
 \end{array}$$

$$\begin{array}{r}
 -10110 \\
 \underline{10010} \\
 00100
 \end{array}$$

$$\begin{array}{r}
 -10110 \\
 \underline{10011} \\
 00011
 \end{array}$$

1's complement

logic operation

$$(1010111)_2$$

1's complement is inverse of digits

$$(01010000)_2 \text{ 1's comp.}$$

$$(01010001)_2 \text{ 2's complement}$$

$$(0101)_2 = (5)_{10}$$

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$(r-1)$'s r 's complements

radix = base

$$r=10 \quad L=4 \quad N=2468 \quad r^4=10000$$

$$r^4 - 1 = 9999$$

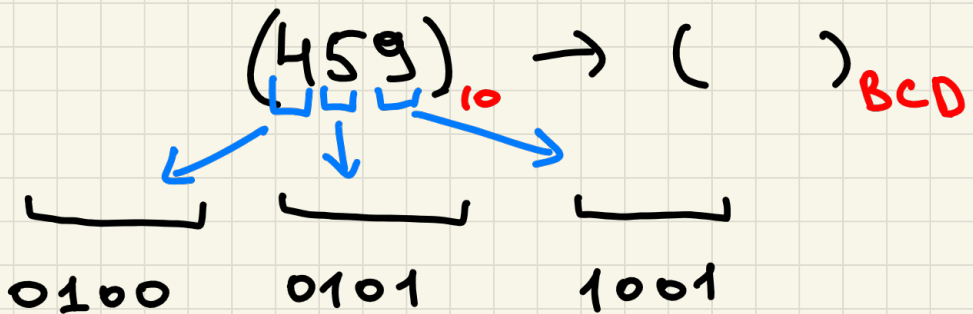
Therefore

$$\begin{array}{r} 9999 \\ - 2468 \\ \hline \underline{7531} \end{array}$$

Decimal $(185)_{10}$

Binary $(10111001)_2$

Binary Coded Decimal BCD



$$4_{10} + 8_{10} = 12_{10}$$

BCD range
(0-9)

$$0100_2 + 1000_2 = 1100_2 \rightarrow 12_{10}$$

$$0110_2 \rightarrow 6_{10}$$

$(\underline{0001} \underline{0010})_{BCD}$

$$\begin{array}{r}
 + 162 \\
 769 \\
 \hline
 931
 \end{array}$$

Binary Coded Decimal

$$\begin{array}{r}
 0001 \quad 0110 \quad 0010 \\
 0111 \quad 0110 \quad 1001 \\
 \hline
 \end{array}$$

$$\begin{array}{r}
 1000 \quad \underline{1100} \quad \underline{1011} \\
 \text{doesn't exist in BCD} \quad \text{more than 9} \\
 \text{to solve add } 6_{10}
 \end{array}$$

$$\begin{array}{r}
 0110 \quad 0110 \\
 \hline
 \underline{1001} \quad \underline{0011} \quad \underline{0001} \\
 9 \quad 3 \quad 1
 \end{array}$$

encoding $\rightarrow$ public UTF-8
 encryption $\rightarrow$ special key

BOOLEAN LOGIC

$$A+B \rightarrow A \text{ or } B$$

$$A.B \rightarrow A \text{ and } B$$

$$A', \bar{A} \rightarrow \text{NOT } A$$

inputs

A	B	A.B	A+B	A'	B'
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0

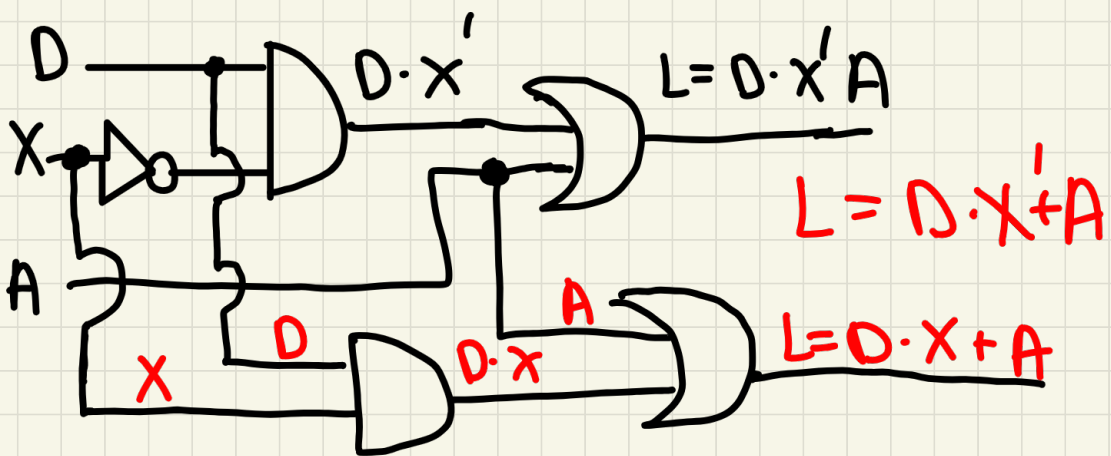
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$$L = \underbrace{D \cdot X'}_{\substack{\uparrow \\ \text{output}}} + \underbrace{A}_{\substack{\uparrow \\ \text{inputs}}}$$

D	X	A	X'	D · X'	D · X' + A
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0

output

0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1



A	B	C	AB	A'C	BC		AB + A'C
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	1	0	1	1	1
1	0	0	0	0	0	0	0

1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

$$F(A, B, C) = (A+B)(A+C)$$

$$AA + AC + AB + BC \quad \text{Distributivity}$$

$$A + AC + AB + BC \quad A = AA$$

$$A(\underbrace{1 + C + B}_1) + BC$$

$$A \cdot 1 + BC = A + BC$$

Sum of Products SOP

$$F(A, B, C) = A'B'C + ABC + A'BC'$$

Product of Sum

$$\begin{aligned}
 F(A,B,C) &= (A+B)(A+C) \\
 &= (A+A) + A+C + AB+BC \\
 &= A + A + C + AB+BC \\
 &= A(1+C) + BC \\
 &= A \cdot 1 + BC = A+BC
 \end{aligned}$$

$$\begin{aligned}
 F &= A + BC + EF \\
 &\text{Sum of Product terms} \\
 G &= (A+B)(C+D+E)(F+H) \\
 &\text{Product of Sum terms}
 \end{aligned}$$

(k)	D	X	A	X'	D X	D X' + A	
0	0	0	0	1	0	0	1 x
1	0	0	1	0	0	1	0 x'
2	0	1	0	0	0	0	
3	0	1	1	0	0	1	
4	1	0	0	1	1	1	D' x A
5	1	0	1	0	1	1	D' x A
6	1	1	0	0	0	0	D x A
7	1	1	1	0	0	1	D x A

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Q 1

$$y + x'z + xy' = x + y + z$$

$$y + xy' + x'z$$

$$(y+x)(y+y') + x'z$$

$$y + x + x'z$$

$$y + (x+x')(x+z)$$

$$y + x + z = x + y + z$$

Q₂

$$x'y' + x'y + xy = x'y$$

$$x'(y' + y) + xy$$

$$x' + xy$$

$$(x' + x)(x' + y)$$

$$1 \cdot x' + y = x'y$$

Q₃

$$x'y' + y'z + xz + xy + yz' \quad 13 \text{ operations}$$

$$x'y' + y'z(x + x') + xz + xy + yz'$$

$$\underline{x'y'} + xy'z + \underline{x'y'}z + xz + xy + yz'$$

$$x'y'(1 + z) + xy'z + xz + xy + yz'$$

$$x'y' + \underline{xy'z} + \underline{xz} + xy + yz'$$

$$x'y' + xz(y' + 1) + xy + yz'$$

$$x'y' + xz + xy + yz'$$

$$x'y' + xz + xy(z+z') + yz'$$

$$x'y' + xz + xyz + xyz' + yz'$$

$$x'y' + xz + xyz + yz'(x+1)$$

$$x'y' + \underline{xz} + \underline{xyz} + yz'$$

$$x'y' + xz(1+z) + yz'$$

$$x'y' + xz + yz'$$

8 operations

SOP

$$F(a,b,c) = a + bc + abc'$$

$$A(a,b,c) = \underbrace{abc}_{\text{Minterms}} + \underbrace{a'bc}_{\text{Minterms}} + \underbrace{abc'}_{\text{Minterms}} \quad \text{Canonical (product terms)}$$

K						Minterms
	a	b	c	A	A	SOP
0	0	0	0	$a'b'c'$	0	$x=1$ $x'=0$
1	0	0	1	$a'b'c$	0	
2	0	1	0	$a'bc'$	0	
3	0	1	1	$a'bc$	1	
4	1	0	0	$ab'c'$	0	
5	1	0	1	$ab'c$	0	
6	1	1	0	abc'	1	
7	1	1	1	abc	1	

$$A(a,b,c) = \sum m(3,6,7) \rightarrow \text{minterms}$$

↓
sum of products

POS (product of sum)

$$G(a,b,c) = \underline{(a+b+c')} \cdot \underline{(a'b'+c')} \cdot \underline{(a+b+c)}$$

Max terms:

$$x=0$$

POS

$$x'=1$$

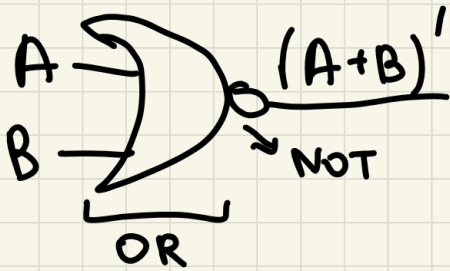
	a	b	c	P	
0	0	0	0	abc	1
1	0	0	1	abc'	1
2	0	1	0	$ab'c$	0
3	0	1	1	$ab'c'$	0
4	1	0	0	$a'b c$	0
5	1	0	1	$a'b c'$	0
6	1	1	0	$a'b' c$	0
7	1	1	1	$a'b' c'$	1

$$P(a,b,c) = \underbrace{\Pi M}_{\text{POS}} (\underbrace{0, 1, 7}_{\text{max terms}})$$

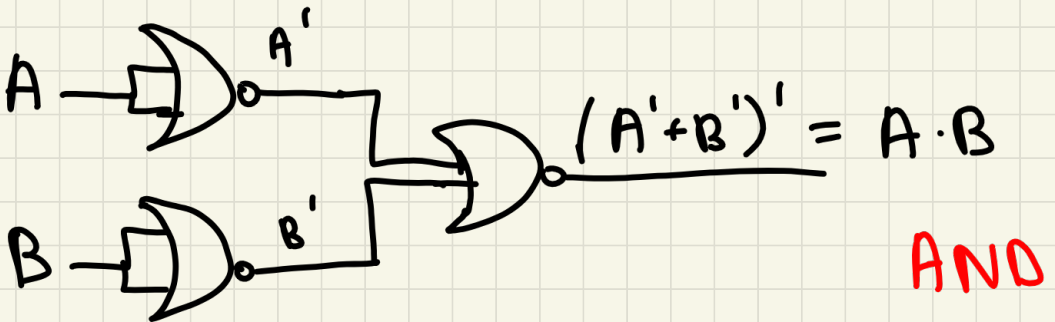
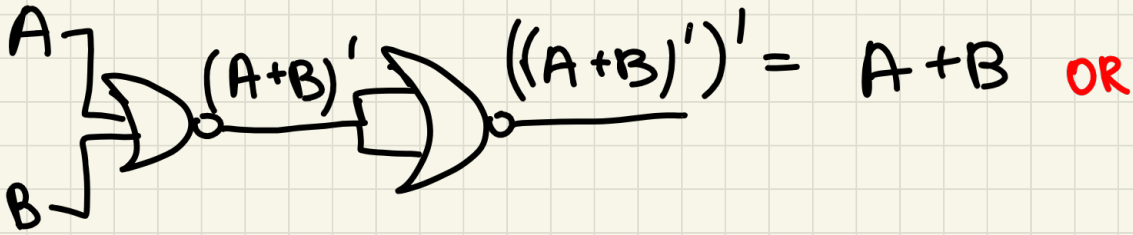
Always write Maxterms POS $x'=1$
 Minterms SOP $x=1$
 $x=0$
 $x'=0$

January 22

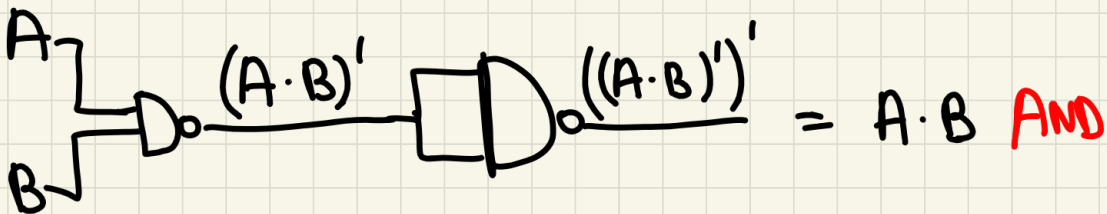
NOR Gate - Universal Gate means
 Function completeness (OR, AND, NOT)



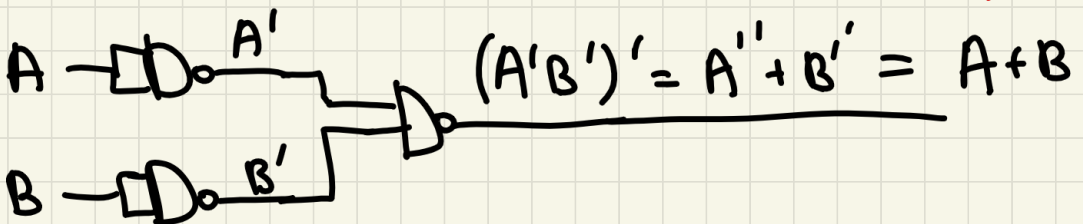
NOR



AND



OR



$$F(a, b, c) = a'b'c + bc' + ac' \quad (\text{Standard})$$

↓ missing a
↓ missing b

$$F(a, b, c) = a'b'c + \cancel{abc'} + a'bc' + \cancel{abc'} + \underline{ab'c'}$$

$$F(a, b, c) = a'b'c + a'bc' + abc' + ab'c'$$

SOP Canonical form

$$F(a, b, c) = (a+b+c) \cdot (a+b) \cdot (a+c') \quad (\text{Standard})$$

↓ needs c
↓ needs b

$$\cancel{(a+b+c)} \cdot \underline{(a+b+c)} \cdot \cancel{(a+b+c')} \cdot \underline{(a+b+c')} \cdot (a+b'+c')$$

$$(a+b+c) (a+b+c') (a+b'+c')$$

POS Canonical form

1-Variable map

x	F(x)
0	F(0)
1	F(1)

2 variable maps

x	y	F(x)
0	0	F(0)
0	1	F(1)
1	0	F(2)
1	1	F(3)

	0	1
0	00	01
1	10	11

2 variable maps

x	y	z	F(x)
0	0	0	F(0)
0	0	1	F(1)
0	1	0	F(2)
0	1	1	F(3)
1	0	0	F(4)
1	0	1	F(5)
1	1	0	F(6)
1	1	1	F(7)

x \ yz	00	01	11	10
0	$x'z'y'$ 000 0	$x'y'z$ 001 1	$x'yz$ 011 3	$x'yz'$ 010 2
1	xyz 100 4	$xy'z$ 101 5	$x'yz$ 111 7	$x'yz'$ 110 6

→ gray codes

SOP | $x=1$
 $x'=0$

$$F(x,y,z) = \sum m(1,3,5,7)$$

$$x'y'(z+z')$$

$$x'y'(1) = x'y'$$

0	0 ₀	1 ₁	1 ₃	0 ₂
1	0 ₄	1 ₅	1 ₇	0 ₆

Group of 4

$$(2^n) = 1, 2, 4, 8, 16$$

$$x' y' z + x' y z + x y z' + x y z$$

$$z (x' y' + x' y + x y' + x y)$$

$$z (x' \underbrace{(y' + y)}_1 + x \underbrace{(y' + y)}_1)$$

$$z (\underbrace{x' + x}_1)$$

z

Shorter way to calculate

4 variable map

w x y z | f(w, x, y, z)

16 rows

w \ x \ yz	00	01	11	10
00	0000 0	0001 1	0011 3	0010 2
01	0100 4	0101 5	0111 7	0110 6
11	1100 12	1101 13	1111 15	1110 14
10	1000 8	1001 9	1011 11	1010 10

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$$F(a, b, c) = \underbrace{a \cdot b \cdot c'}_{\text{and}} + \underbrace{ab}_{\text{and}} + c$$

OR OR

x

	1	1	
	1	1	

$$SOP = x'y + z$$

② implicant

a

	bc	$b'c'$	$b'c$	bc'
a'	00	01	11	10
a'	0	0	1	1
a	1	0	1	0

$ab'c'$

$a'c$

bc

$$F(a, b, c) = \sum m(1, 3, 4, 7)$$

$$F(a, b, c) = ab'c' + a'c + bc$$

a

	bc	b'c'	b'c	bc'
a'	1	1	1	1
a	0	1	1	0

$$a'(b'c' + b'c + bc + bc') = a'$$

$$a' + c$$

wx \ yz	yz			
	y'z'	y'z	yz	yz'
w'x'	1 ₀	0 ₁	0 ₃	1 ₂
w'x	1 ₄	1 ₅ *	0 ₇	0 ₆
wx	1 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
wx'	1 ₈	0 ₉	1 ₁₀	1 ₁₁

Grouping annotations: Blue outlines group (0,2,8,10) and (4,6,12,14). Green outlines group (5,13) and (1,3,9,11). Purple arrows point to prime implicants 1, 2, 3, 4, and 5.

Grouping:

groupsize 8,4,2,1

$$F(w, x, y, z) = (0, 2, 4, 5, 8, 10, 11, 12, 13, 15)$$

$$F(w, x, y, z) = 1, 2, 3, 4, 5$$

1	0	0	1
1	1	0	0
1	1	1	0
1	0	1	1

Red outlines group (0,2,8,10), (4,6,12,14), (5,13), and (1,3,9,11).

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February 3, 2026

$$F(A, B, C, D) = \sum m(1, 3, 4, 5, 9, 10, 11, 12, 13, 14, 15)$$

AB \ CD				
	00	01	11	10
00		1	1	
01	1	1		
11	1	1	1	1
10		1	1	1

$$F(A, B, C, D) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$$

AB \ CD				
	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

Prime : $BD, A'C'D, A'BC, ABc', ACD$

Combinational Logic Gates

Half adder

$$\begin{array}{r} A \\ + B \\ \hline \text{Sum} \end{array}$$

2 inputs : A, B

2 outputs : S, C

inputs
↔

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$S = A'B + AB'$$

$$C = AB$$

